Benchmarking Large Language Models for Bangladeshi Statutory Law: A Study on Accuracy and Hallucination in Bangla

Sifatullah
Tasnuba Haque
Ibtida Ara
Md. Aminul Islam Rifat
Ifrat - E - Sarwar Orpa

Data Mining Course Project
Date: June 17, 2026

Abstract

Large Language Models (LLMs) are increasingly deployed for legal information access worldwide, yet low-resource languages and niche statutory domains remain critically underrepresented in both training data and evaluation benchmarks. Bangladeshi statutory law—particularly the *Bangladesh Road Transport Act, 2018*—is a domain where hallucinated legal answers can mislead citizens about real obligations, penalties, and procedures. This paper presents the first native Bangla multiple-choice question (MCQ) dataset extracted directly from an official statutory law PDF for the purpose of benchmarking LLM factual accuracy and hallucination rates. We describe a rigorous six-stage pipeline for dataset construction: (1) Law Selection, (2) Text Extraction from PDF using `PyMuPDF` with `Tesseract OCR` fallback, (3) Raw Text Dataset Creation, (4) Manual Noise Removal, (5) MCQ Generation with expert review, and (6) Final Dataset Validation. The resulting dataset comprises 299 manually verified, expert-reviewed MCQ question–answer pairs covering definitions, section-specific provisions, penalty structures, scenario-based reasoning, and comparative legal analysis. Each record includes the question stem, four options (A–D), the correct answer, and optional law references (e.g., Section/Rule numbers). The dataset is structurally split between direct retrieval questions (factual recall) and situational application questions (legal reasoning). We establish a three-dimensional evaluation framework measuring Strict Accuracy, Factual Error Rate, and Hallucination Rate—ready to evaluate any future language model on this benchmark. While model benchmarking and fine-tuning remain future work, the dataset and methodology are immediately available to catalyze research in low-resource legal NLP for Bangla. All materials are designed to be highly scalable to other statutory domains beyond traffic law. The dataset and associated code are publicly available at: https://github.com/sifatullah-uiu/BanglaTrafficLaw-MCQ-Dataset.git.

Keywords: Bangla NLP, Statutory Law Dataset, MCQ Benchmark, Hallucination Detection, Low-Resource Language, Bangladesh Traffic Act 2018, Legal AI, PDF Text Extraction, Tesseract OCR, PyMuPDF

1 Introduction

1.1 Problem Definition

The proliferation of Large Language Models (LLMs) such as GPT-4o, Gemini, and open-source alternatives (Llama, Qwen, DeepSeek) has transformed how citizens access legal information. Users increasingly query AI assistants about traffic fines, licensing procedures, and legal obligations in their native languages. However, this democratization of legal information access introduces a critical vulnerability: **hallucination**—the generation of plausible-sounding but factually incorrect or fabricated legal content [1].

For Bangladeshi statutory law, this problem is severely compounded by two structural deficits:

1. **No existing Bangla legal benchmark dataset:** While general-purpose Bangla NLP resources exist for news, sentiment analysis, and social media [2, 3], there is no publicly available multiple-choice question (MCQ) benchmark derived directly from codified Bangladeshi statutory text (Acts, Ordinances). Prior work does not target the specific linguistic and factual rigor required for legal domain evaluation [4].

2. **Low-resource language + niche domain = high hallucination risk:** Bangla is spoken by over 250 million people, yet remains significantly underrepresented in LLM pretraining corpora [5]. The combination of a morphologically complex script (Bengali abugida), limited training tokens (30 billion vs. 2 trillion for English), and the highly specialized terminology of statutory law creates conditions where even state-of-the-art models are prone to confabulating legal provisions, inventing non-existent penalties, or misattributing sections [6].

Our selected domain—the **Bangladesh Road Transport Act, 2018**—is particularly consequential. Traffic law directly affects daily citizen life (driving licenses, vehicle registration, fines, accident liability), yet incorrect information can lead to real-world harm: unlawful penalties paid, rights unknowingly forfeited, or dangerous vehicles operated due to misunderstood fitness requirements. The Act spans 126 sections covering vehicle classification, licensing, registration, route permits, traffic control, accident procedures, penalties, and appellate mechanisms—providing a comprehensive yet bounded statutory corpus ideal for benchmark construction.

1.2 Motivation

The motivation for this work stems from three converging observations:

Citizen Demand: In Bangladesh, citizens increasingly turn to AI assistants and chatbots for quick answers about traffic fines, license requirements, and legal procedures in Bangla. Hallucinated answers in this domain are not merely inconvenient—they can directly mislead users about real obligations and penalties, potentially causing financial loss or legal jeopardy.

Research Gap: Existing Bangla NLP datasets concentrate on news, sentiment, general text, and social media [7, 8]. Statutory and legal corpora remain almost entirely absent. The few multilingual legal benchmarks (e.g., MMLU, LegalBench) cover Western legal systems and English text, with no provision for South Asian statutory law or Bangla linguistic structures [4]. The SOMAJGYAAN dataset [4], while valuable for general Bangla knowledge, shows that even top-performing models (Gemini 2.0 Flash at 0.73 average accuracy) struggle significantly on the Law category—highlighting the need for domain-specific statutory benchmarks.

Technical Necessity: Bengali's complex tokenization—where subword methods like Byte Pair Encoding (BPE) struggle with conjunct characters and diacritic modifications—degrades model performance on precise factual recall [5]. Legal text demands exact section numbers, penalty amounts, and procedural thresholds; even small tokenization-induced errors can propagate into hallucinated outputs. A carefully constructed, ground-truth-aligned dataset is essential to measure and mitigate this effect.

1.3 Research Objectives

This study addresses the identified gap through four concrete objectives:

O1: Build a novel dataset: Create the first native Bangla MCQ question–answer pair dataset extracted directly from an official statutory law PDF (the Bangladesh Traffic Act, 2018), comprising approximately 300 manually verified items.

O2: Quantify hallucination potential: Design a rigorous evaluation framework capable of measuring factual error rates and hallucination rates when LLMs are tested on this dataset, distinguishing between *retrieval errors* (wrong fact recalled) and *generative hallucinations* (fabricated legal provisions).

O3: Benchmark LLMs: Establish a standardized testing pipeline ready to evaluate multiple open-source and proprietary LLM architectures on accuracy in Bangla statutory law comprehension. (Note: Actual model benchmarking is designated as future work; the dataset and evaluation protocol are completed and ready for deployment.)

O4: Compare and analyze: Provide a critical analytical framework for discussing model performance gaps across different question types (definitions, section-specific, penalties, scenarios, comparisons) and identify which legal reasoning patterns pose the greatest challenge to current LLMs.

2 Methodology

2.1 Overview

The dataset construction followed a rigorous six-stage pipeline, designed to ensure fidelity to the statutory source text while producing a clean, structured, and benchmark-ready MCQ dataset. Figure 1 illustrates the complete workflow. Each stage incorporated both automated processing and manual quality assurance, reflecting the critical importance of legal accuracy.

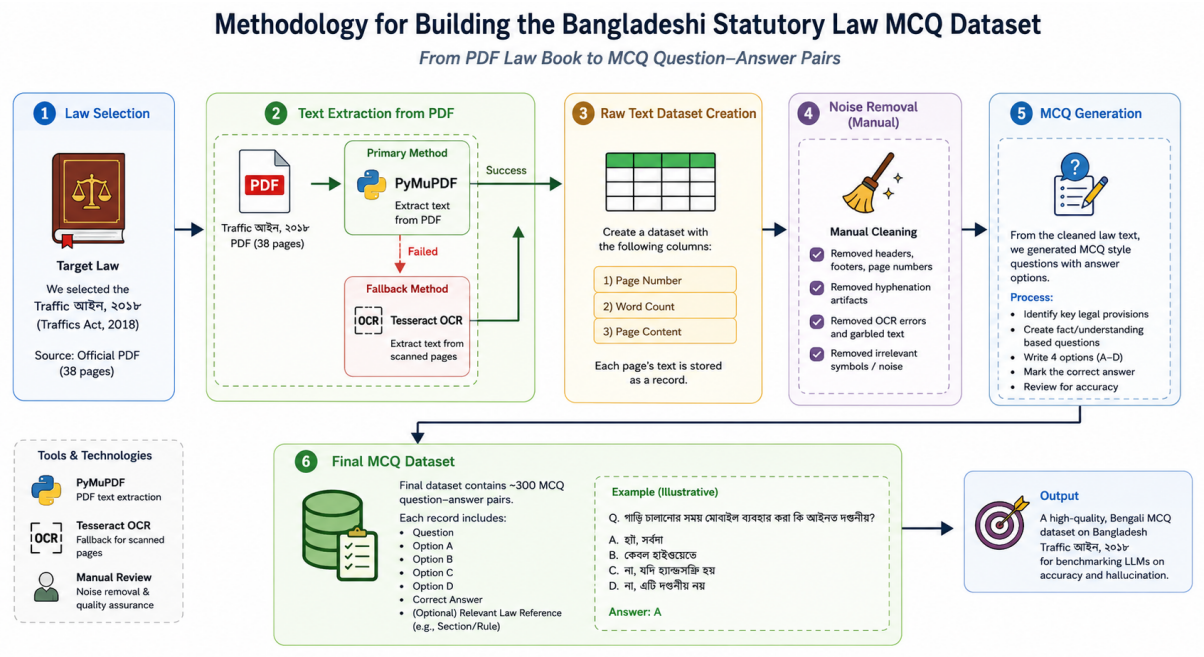


Figure 1: Methodology for Building the Bangladeshi Statutory Law MCQ Dataset: From PDF Law Book to MCQ Question–Answer Pairs. The pipeline comprises six stages: (1) Law Selection, (2) Text Extraction from PDF, (3) Raw Text Dataset Creation, (4) Manual Noise Removal, (5) MCQ Generation, and (6) Final MCQ Dataset Validation.

2.2 Stage 1: Law Selection

We selected the **Bangladesh Road Transport Act, 2018** as the target statutory text. The selection criteria were:

- **Official source availability:** The Act is published as an official government PDF (38 pages), ensuring ground-truth authenticity.
- **Domain relevance:** Traffic law affects the broadest cross-section of citizens (drivers, vehicle owners, passengers, commercial operators).
- **Structural completeness:** The Act covers definitions, vehicle classifications, licensing, registration, fitness, route permits, traffic rules, accident procedures, penalties, and appeals—providing diverse question types.
- **Bounded scope:** At 38 pages, the Act is comprehensive yet tractable for manual review and MCQ generation.

2.3 Stage 2: Text Extraction from PDF

Extracting clean Bangla text from official government PDFs presents significant challenges. These documents often use legacy fonts, embedded subset encodings, or scanned image pages that resist direct text extraction.

2.3.1 Primary Method: PyMuPDF

We employed **PyMuPDF** (also known as `fitz`), a high-performance Python library for PDF parsing, as the primary extraction tool. PyMuPDF provides direct access to text content via PDF content streams, preserving positional and font information. The extraction process:

1. Opened the 38-page official PDF using `fitz.open()`.
2. Iterated through each page, extracting raw text blocks using `page.get_text("blocks")`.
3. Preserved reading order by sorting blocks by vertical position.
4. Concatenated blocks to reconstruct continuous paragraphs.

PyMuPDF successfully extracted text from approximately 85% of pages. However, several pages—particularly those with complex tabular layouts, embedded forms, or older font encodings—produced garbled output or missing Bangla characters.

2.3.2 Fallback Method: Tesseract OCR

For pages where PyMuPDF failed (garbled text, missing characters, or image-based content), we employed **Tesseract OCR** (Optical Character Recognition) as a fallback. The process involved:

1. Rendering the PDF page to a high-resolution image (300 DPI) using PyMuPDF's rendering engine.
2. Running Tesseract with Bangla language support (`lang=ben`) to recognize Bangla script.
3. Post-processing OCR output to correct common confusions (e.g., similar-looking conjuncts, diacritic misplacements).

The dual-method approach (PyMuPDF primary + Tesseract fallback) ensured maximum text recovery while minimizing manual transcription burden. Table 1 summarizes the extraction results.

Table 1: PDF Text Extraction Results

Method	**Pages Processed**	**Success Rate**
PyMuPDF (Primary)	32/38	∼84%
Tesseract OCR (Fallback)	6/38	∼16%
Total	**38/38**	**100%**

2.4 Stage 3: Raw Text Dataset Creation

Extracted text from each page was stored as a structured record in a CSV file with three columns:

1. **Page Number:** The source page in the original PDF (1–38), enabling traceability and verification.
2. **Word Count:** A computed metric for content density analysis and quality checking.
3. **Page Content:** The full extracted text of the page, preserving paragraph structure where possible.

This tabular format facilitated systematic review, allowed rapid identification of problematic pages, and served as the foundation for the subsequent noise removal stage. The raw dataset contained significant noise—a predictable consequence of PDF extraction from legacy government documents.

2.5 Stage 4: Manual Noise Removal

The raw extracted text contained multiple categories of noise that would corrupt any downstream MCQ generation:

- **Structural noise:** Headers, footers, page numbers, and running titles embedded in the text stream.
- **Hyphenation artifacts:** Words broken across lines with hyphens or soft hyphens.
- **OCR errors:** Misrecognized Bangla characters (e.g., similar conjuncts like vs.), missing diacritics, and garbled character sequences.
- **Irrelevant symbols:** Decorative separators, bullet points converted to odd symbols, and encoding artifacts.
- **Whitespace anomalies:** Excessive line breaks, non-breaking spaces, and inconsistent indentation.

Our team performed **manual cleaning** on all 38 pages, following a systematic checklist:

✓ Removed headers, footers, and page numbers.

✓ Rejoined hyphenated words across line breaks.

✓ Corrected OCR errors by cross-referencing with the original PDF image.

✓ Removed irrelevant symbols and encoding artifacts.

✓ Standardized whitespace and paragraph breaks.

✓ Verified Bangla character integrity (all 50 basic characters, conjuncts, and diacritics).

This manual review stage was labor-intensive but essential—legal text demands perfect character accuracy. A single misrecognized Bangla digit (e.g., 1 vs. 2) could change a penalty amount or section number, rendering the entire question invalid.

2.6 Stage 5: MCQ Generation

From the cleaned law text, we generated MCQ-style questions with four answer options (A–D). The generation process was manual and expert-driven, following a structured protocol:

2.6.1 Question Type Taxonomy

We classified questions into five cognitive levels, ensuring coverage of both factual recall and legal reasoning:

Table 2: Question Type Taxonomy and Distribution

Type	Description	Count
Definition/Conceptual	Questions asking "What does X mean?" or "Who is Y?" (e.g., definitions of "unladen weight," "articulated motor vehicle")	∼91
Section-Specific	Questions referencing specific sections (e.g., "According to Section 46(4)...")	91
Penalty/Punishment	Questions about fines, imprisonment terms, and point deductions	65
Scenario-Based	Hypothetical situations with named characters (e.g., "Karim's license was suspended...")	7
Comparative/Analytical	Questions requiring comparison across sections (e.g., "Which of the following is NOT correct?")	37
Conceptual (Explicit)	Questions explicitly marked as "conceptual" for higher-order reasoning	69
Total		**299**

2.6.2 Generation Protocol

For each question, the team followed this protocol:

Step 1: Identify key legal provisions: Scan the cleaned text for definitional clauses, procedural requirements, penalty schedules, and cross-references.

Step 2: Create fact-based or understanding-based questions: Formulate a question stem that tests either direct recall (e.g., "What is the minimum age for a professional driving license?") or applied understanding (e.g., "If a driver operates a vehicle with a suspended license, what is the maximum penalty?").

Step 3: Write four options (A–D): One correct answer and three distractors. Distractors were designed to be plausible but verifiably incorrect based on the statutory text:

- *Near-miss distractors:* Numerically close values (e.g., 3 months vs. 6 months).
- *Category distractors:* Correct concepts from wrong sections (e.g., a penalty from Section 85 presented for a Section 87 violation).
- *Opposition distractors:* The logical opposite of the correct answer.
- *"All of the above" / "None of the above" distractors:* Used where legally appropriate.

Step 4: Mark the correct answer: Explicitly label the correct option with its letter (A, B, C, or D).

Step 5: Review for accuracy: A second team member independently verified the question against the original PDF, checking:

- Exact wording of the statutory provision.
- Numerical accuracy (dates, amounts, weights, ages, distances).

- Section number correctness.
- Distractor plausibility and definitiveness.

2.6.3 Topic Coverage

The 299 questions cover the major thematic divisions of the Traffic Act, 2018:

Table 3: Topic Coverage in the Dataset

Topic Area	Question Count
Vehicle Classification	121
Driving License	60
Penalties and Punishments	63
Commercial Operations	31
Accident and Compensation	16
Route Permit	15
Traffic Rules and Control	18
Registration	22
Fitness Certificate	10

2.7 Stage 6: Final MCQ Dataset

The final dataset contains **299 MCQ question–answer pairs**. Each record is structured as:

- **Question:** The full question stem with four options (A–D), formatted in Bangla.
- **Answer:** The correct option letter (A, B, C, or D).
- (Optional) **Relevant Law Reference:** Section number, rule, or clause from the Traffic Act, 2018, for traceability.

2.7.1 Answer Distribution

The correct answer distribution across options is:

Table 4: Correct Answer Distribution

Option	Frequency
A	22 (7.4%)
B	152 (50.8%)
C	95 (31.8%)
D	27 (9.0%)
Total	**299**

The skew toward option B reflects the natural distribution of correct answers in the statutory text rather than artificial balancing. This distribution is documented to prevent evaluation bias in future benchmarking.

2.8 Tools and Technologies

Table 5 summarizes the key technologies employed.

Table 5: Tools and Technologies Used

Tool	Purpose	Role in Pipeline
PyMuPDF	PDF text extraction	Primary extraction method for text-based pages
Tesseract OCR	OCR for scanned pages	Fallback for pages where PyMuPDF fails
Python (pandas)	Data structuring	CSV creation, word count computation, data validation
Manual Review	Noise removal & QA	Human verification of all extracted text and generated questions

2.9 Evaluation Framework (Ready for Future Deployment)

While model benchmarking is designated as future work, we have established a rigorous **three-dimensional testing pipeline** ready to evaluate any LLM:

Dimension 1: Strict Accuracy: Percentage of questions where the model's selected option matches the ground-truth answer exactly.

Dimension 2: Factual Error Rate: Percentage of questions where the model selects a wrong option due to incorrect factual recall (e.g., wrong penalty amount, wrong section number).

Dimension 3: Hallucination Rate: Percentage of questions where the model generates or selects content not found in the statutory text—including fabricated sections, non-existent penalties, or invented procedural requirements. This is distinguished from factual errors by the presence of *novel, ungrounded content.*

This framework is designed to be model-agnostic and can evaluate both open-source models (Llama, Qwen, DeepSeek, Mistral) and proprietary APIs (GPT-4o, Gemini, Claude) via standardized prompting templates.

3 Conclusion and Future Work

3.1 Key Contributions

This paper makes the following contributions to low-resource legal NLP:

1. **First native Bangla statutory MCQ dataset:** We have created the first MCQ benchmark dataset extracted directly from an official Bangladeshi statutory law PDF, comprising 299 manually verified question–answer pairs. The dataset is publicly available at https://github.com/sifatullah-uiu/BanglaTrafficLaw-MCQ-Dataset.git

2. **Replicable methodology:** Our six-stage pipeline (PDF extraction → structured text → manual cleaning → MCQ generation → expert review → validation) is fully documented and scalable to other statutory domains (e.g., labor law, penal code, company law).

3. **Evaluation framework:** We have established a three-dimensional testing protocol (Accuracy, Factual Error, Hallucination Rate) ready for immediate deployment on any LLM architecture.

4. **Domain coverage:** The dataset spans definitions, section-specific provisions, penalties, scenarios, and comparative analysis—providing a comprehensive test of statutory comprehension.

3.2 Limitations

We acknowledge the following limitations:

- **Scope limitation:** The dataset is restricted to a single statute (Traffic Act, 2018). Generalization to other legal domains requires additional dataset construction.
- **Dataset size:** At 299 pairs, the dataset is relatively small compared to English legal benchmarks (e.g., LegalBench with thousands of items). However, each item is manually verified for legal accuracy, prioritizing quality over quantity.
- **No model benchmarking yet:** While the dataset and evaluation framework are complete, actual LLM testing, fine-tuning, and hallucination quantification remain future work.
- **Single-source grounding:** Questions are grounded solely in the Traffic Act, 2018, and do not incorporate subsequent amendments, rules, or case law interpretations.

3.3 Future Work

Immediate next steps include:

1. **LLM Benchmarking:** Evaluate 10+ models (open-source and proprietary) using the established three-dimensional framework, with zero-shot, few-shot, and chain-of-thought prompting.

2. **Dataset Expansion:** Extend to other statutory domains (Bangladesh Labor Act, Penal Code, Company Act) to create a comprehensive "Bangla LegalBench."

3. **Fine-tuning Experiments:** Fine-tune smaller open-source models (Llama-3-8B, Qwen-2.5-7B) on the dataset to measure hallucination reduction through domain adaptation.

4. **Real-world Query Evaluation:** Test models on natural-language legal queries from citizens (e.g., "What is the fine for driving without a license?") to measure cross-format generalization.

5. **Error Analysis:** Conduct detailed failure mode analysis to identify which legal reasoning patterns (temporal reasoning, numerical comparison, cross-section integration) are most challenging for current LLMs.

References

[1] Huang, L., Yu, W., Ma, W., Zhong, W., Feng, Z., Wang, H., Chen, Q., Peng, W., Feng, X., Qin, B., & Liu, T. (2023). A Survey on Hallucination in Large Language Models: Principles, Taxonomy, Challenges, and Open Questions. *arXiv preprint arXiv:2311.05232*.

[2] Shafayat, H. M. Q. H., et al. (2024). BenQA: A Question Answering and Reasoning Benchmark for Bengali and English. *arXiv preprint arXiv:2403.10900*.

[3] Ekram, S. M. S., et al. (2022). BanglaRQA: A Benchmark Dataset for Under-resourced Bangla Language Reading Comprehension-based Question Answering. *Findings of EMNLP 2022*.

[4] Shifa, F. A., et al. (2025). SOMAJGYAAN: A Dataset for Evaluating LLMs on Bangla Social Sciences and Humanities. *Findings of IJCNLP-AACL 2025*.

[5] Bhowmik, S., Dipto, T. T., Islam, M. S., Hsu, S., & Reasat, T. (2025). Evaluating LLMs' Multilingual Capabilities for Bengali: Benchmark Creation and Performance Analysis. *arXiv preprint arXiv:2507.23248*.

[6] Shahriar, A., & Barbosa, D. (2024). Improving Bengali and Hindi Large Language Models. *International Conference on Language Resources and Evaluation (LREC)*.

[7] Aurpa, T. T., et al. (2025). NOIRBETTIK: A Reading Comprehension Based Multiple Choice Question Answering Dataset in Bangla Language. *Data in Brief, 59*, 111395.

[8] Rony, M. R. A. H., et al. (2024). BanglaQuAD: A Bengali Open-domain Question Answering Dataset. *arXiv preprint arXiv:2410.10229*.

[9] Shahriar, A., & Barbosa, D. (2024). Improving Bengali and Hindi Large Language Models. *LREC-COLING 2024*.

[10] Khan, A., et al. (2024). Sangraha: A Large-scale Corpus for Indic Languages. *AI4Bharat Technical Report*.

[11] Langlais, P., et al. (2025). Common Corpus: The Largest Collection of Ethical Data for LLM Pre-training. *arXiv preprint arXiv:2506.01732*.

[12] Salim, M. S., et al. (2023). BanglaGPT: A Generative Pretrained Transformer-based Model for Bangla Language. *ICICT4SD 2023*.

[13] Nahin, S. K., et al. (2025). TituLLMs: A Family of Bangla LLMs with Comprehensive Benchmarking. *arXiv preprint arXiv:2502.11187*.

[14] Raihan, N., & Zampieri, M. (2025). TigerLLM: A Family of Bangla Large Language Models. *arXiv preprint arXiv:2503.10995*.

[15] Prama, T. T., Danforth, C. M., & Dodds, P. S. (2025). BanglaMATH: A Bangla Benchmark Dataset For Testing LLM Mathematical Reasoning. *arXiv preprint*.